



Selection of Significant Features for Snow Avalanche Forecasting

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Abstract

Avalanche forecasting is most widely practiced approach for avalanche hazard mitigation in snow bound mountainous regions. In this context, machine learning techniques are typically used to discriminate between avalanche or no-avalanche scenarios. However, limited studies have been focused on selection of significant features that influence avalanche formation. Consequently, prevailing forecasting models often access redundant data leading to slower learning processes, increased computational complexity, and potential overfitting that ultimately compromises its generalization ability. This study proposes adoption of feature selection as a fundamental step in avalanche forecast modeling. In this regard, we demonstrate the usage of a popular feature selection technique SVM-RFE to identify the most relevant feature subset for avalanche forecasting in Bandipore-Gurez region in North-western part of Indian Himalayas. The study explored a set of 40 features (including the features that represented snow-meteorological perturbations occurred over past many days) and identified subsets of 7 to 15 features representing fresh rainfall, fresh snow, cumulative seasonal snow, temperatures, wind speed and direction, and sunshine as the most significant contributors to the avalanche occurrences. Also, some of these selected features represent information up to past 2 to 3 days indicating that the avalanching conditions in the study area ripe over a longer period of time rather than resulting from rapid meteorological changes. Further, the performances of classification models with selected subsets of features either matched or exceeded their respective performances with entire set of 40 features. These findings provide valuable insights for researchers and practitioners in avalanche hazard management and may contribute to development of more efficient forecasting models.

Keywords Avalanche forecasting · Feature selection · Recursive feature elimination · Machine learning · Support vector machine · Random forest

1 Introduction

Avalanche is a natural occurring phenomenon which globally impacts socio-economic activities and human lives in snow bound mountainous terrain. Hence, effective avalanche hazard mitigation is crucial for ensuring stability and safety in these regions. Over the decades, a range of models and systems has been developed for assistance in the decision making process of snow avalanche forecasting varying from expert systems [1] and models using snowpack information [2–4] to a series of machine learning-based tools [5–15]. Of

late, machine learning techniques have gain more popularity globally for avalanche hazard prediction due to their complex data handling capabilities and thereby serving better accuracy. Some of the utilized algorithms are as follows: *k*-nearest neighbor [5, 9], random forest [11, 12], hidden Markov model [7], support vector machine [6, 8, 10, 13], and artificial neural network [7, 16]. Typically, a large number of candidate features are provided to machine learning algorithms for generating a thorough characterization of the classification task [15, 17, 18]. For avalanche forecast modeling as well, input feature sets of a considerable dimension were employed [6, 8, 13, 15, 19, 20]. However, it is often a matter of concern as many features are either irrelevant or redundant [15, 17, 18, 21]. In machine learning, while describing the target concept, irrelevant features do not affect it in anyway and redundant features do not add anything informative to it. Moreover, redundant features possibly add more noise than useful knowledge in describing the concept

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of interest [15, 17, 18]. Such inefficient features, however, highly influence the modeling process leading to overfitting of the classifier as well as slow down the learning process and may also compromise model's generalization abilities [17, 18]. Similar observations were reported in a machine learning-based avalanche susceptibility mapping work also by Tiwari et al. [13] and Cetinkaya and Kocaman [15]. Thus, irrelevant and redundant data causes over consumption of resources and affects generalization ability of machine learning model [13, 15, 21, 22].

From preceding paragraph, it can be inferred that selection of most valuable features from input dataset is an essential initial step in the field of machine learning-based avalanche forecast modeling as well [6, 7, 13, 15, 19, 20]. It improves performance of model by eradicating noisy, irrelevant, and redundant features and lends benefits in terms of time and space complexity of the training algorithm [13, 15, 17, 21, 23].

Feature selection methods can be classified into filter, wrapper, and embedded techniques based on their relationship with learning methods [13, 21, 24]. In filter feature selection method, features are selected based on statistical measures. In other words, filter methods assess the relevance of variables by looking only at the intrinsic properties of the data without taking into account any information provided by the classification algorithm [13, 23–27]. Moreover, it does not consider the correlation between the classifier and features and also fails to recognize the patterns properly during the learning phase [24, 27]. Few of filter feature selection measures are as follows: correlation criteria, mutual information (MI), Relief-F, and signal-to-noise ratio (SNR) [13, 23, 25]. Embedded methods perform feature selection during the modeling algorithm's execution. These methods are thus embedded in the algorithm either as its normal or extended functionality. Common embedded methods include various types of decision tree algorithms: CART, C4.5, random forest, and other algorithms (e.g., multinomial logistic regression and its variants) [25]. According to Venkatesh and Anuradha [27], it is a computationally costlier method and is not suitable for high-dimensional data, due to which it leads to poor model generality. Most of the embedded feature selection methods are based on a somewhat penalization term, i.e., features are penalized depending on their values with some methods explicitly constraining the number of features, while others penalizing the number of features [26]. Moreover, embedded techniques are conceptually complex and modifications to the classification algorithm may leads to a poor performance [24, 28]. Wrapper techniques (such as Boruta, GA, PSO, RF-RFE, and SVM-RFE) wrap around a learning algorithm as the induction algorithm to evaluate the selected feature subset with minimum cross validation error on the training data [13, 23, 24, 26, 27, 29]. A wrapper algorithm is a robust technique and explores the

whole feature space to score feature subsets according to their predictive power [13, 24, 28]. This approach considers the correlation between the features and class labels, as well as accumulates the dependencies between them [26, 27]. Wrapper feature selection method uses the learning algorithm for testing the quality of selected feature subsets, and that is why it generates most accurate results [26, 27].

To the best of our knowledge, study by Pozdnoukhov et al. [6] is the only study that deployed any feature selection method for avalanche forecasting particularly. For avalanche susceptibility mapping, though, Tiwari et al. [13] did demonstrated successful use of a feature selection algorithm. Pozdnoukhov et al. [6] deployed SVM-RFE [29], a wrapper feature selection technique, to select a feature subset for spatio-temporal avalanche forecasting for categorical as well as probabilistic forecasts for Lochaber region of Scotland. However, the main focus of the study was to explore the machine learning method of support vector machines to assess its suitability in avalanche forecasting, and there was no specific detailed analysis conducted on the feature selection. Nonetheless, SVM-RFE has been widely explored in various other domains such as text categorization [23, 30, 31], intrusion detection [32], sentiment analysis [33], and medical diagnosis [29, 30, 34]. The present study explores in detail the applicability of SVM-RFE for identifying significant features in avalanche forecasting. In this regard, we analyzed the data from an avalanche prone area in north-western part of Indian Himalayan region. The results were compared with a previous study and were found to be quite informative. Further, we deployed two machine learning-based classification algorithms, viz. support vector machine (SVM) and random forest (RF), to verify veracity of the results.

Paper is summarized in four subsequent sections. Section 2 describes the data used in the study that also includes the description of the study area. Section 3 elaborates the explored techniques, including feature subset selection algorithm, machine learning classifiers and validation methods. Section 4 describes the overall plan employed for conducting the study. Section 5 explains the results and discussion thereon, and finally, Section 6 concludes the study with remarks on future research scope.

2 Data

The following sub-sections describe the study area and characteristics of data used for this study.

2.1 Study Area

In Indian Himalayas, a major avalanche prone belt runs along Great Himalayan range through the union territories

of Jammu and Kashmir (J&K) and Ladakh and the states of Himachal Pradesh, Uttarakhand, and Sikkim. The present study has been conducted with data from Bandipore-Gurez (BG) sector which is located at the north-western tip of Great Himalayan Range as shown in Fig. 1. A major highway runs in this area along the Kishenganga river in Tulel valley and adjoining Gurez valley. It connects to Bandipore town in Kashmir valley through Razdan pass (3300 m above m.s.l.). In Gurez valley (area on west of Wampore town), about 40 avalanche paths affect the highway stretch of about 25 kms from Jatkushu village to Wampore town. Besides, about 15 avalanche paths affect the lateral tracks. In Tulel valley (area on east of Wampore town), over 100 major and minor avalanche paths affect the highway and lateral tracks. All these 230 representative avalanche paths occupy an area of about 10460 Ha with elevation ranging from 2350 to 4800 m above m.s.l. Distributions of terrain features (slope, aspect, elevation and area) of these avalanche sites are presented in Fig. 2.

According to the scheme proposed by LaChapelle [35], data describing avalanche likelihood is categorized into three classes: Class I, slope stability factors; Class II, snowpack factors; and Class III, meteorological factors. In practice, data describing snowpack (Class II) and meteorological (Class III) factors are generally easier to collect and generalize over a larger region and are more commonly used in avalanche forecasting at the local and regional levels [6]. Defence Geo-informatics Research Establishment (DGRE) operates a surface observatory near Kanzalwan town (2440 m above m.s.l.) in the study area. Various intrinsic snowpack and meteorological features are recorded at this observatory

through automatic and manual devices daily over the day. The logged features include air temperatures (instantaneous, maximum and minimum of the day), fresh snowfall, total snow (aggregated over the season), snowpack height, snow surface temperature, snow surface penetration (due to self-weight of ramsonde guide rod), rainfall, relative humidity, sunshine period, wind speed (instantaneous and averaged), and its direction. From this logged data, we prepared a 40-dimensional dataset comprising of intrinsic as well as derived features (Section 2.2) for the present study. It was augmented with the avalanche observations in the study area (Section 2.3).

Owing to the substantial topographic diversity present across the wide study area, it may be a matter of concern in representing the study area by single observation site. However, focusing on the snow accumulation period supplements the representativeness of a single station. Mid-winter survey conducted by Winstral and Marks [36] noted closer alignment among basin-averaged and single point estimations in contrast with post-accumulation period. Also, coefficient of variation was much low during peak accumulation months, and terrain persuaded unpredictability was likewise less challenging during stable winter circumstances. Similarly, as per the discussions of Wang and Zhang [37], snow stations tend to reflect higher factor representativeness during the stable snow accumulation months. A marked decline in representativeness is observed during the transitional period such as snow melt. By conducting the analysis for the core winter months only (from November of a year to April of the next year), we minimize the inconsistency in representativeness raised by variable melting rates, wind redistribution, and

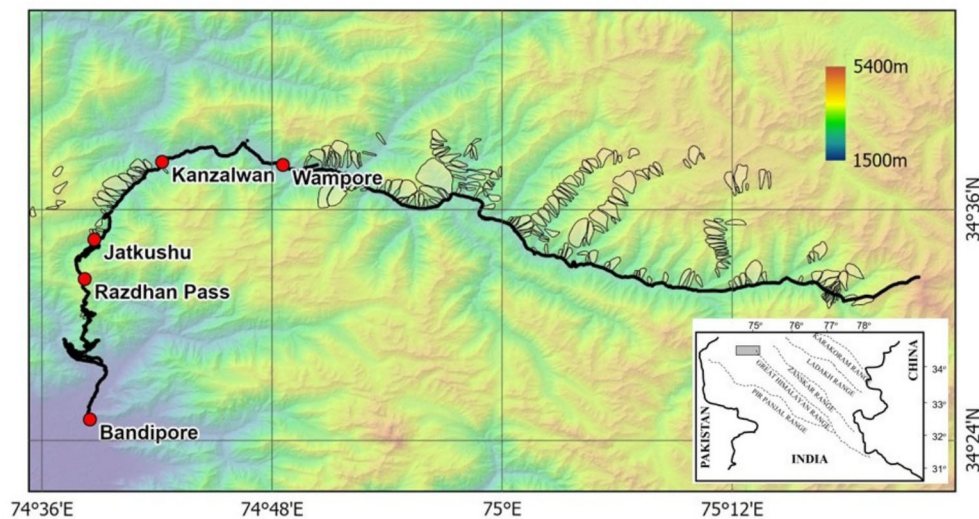
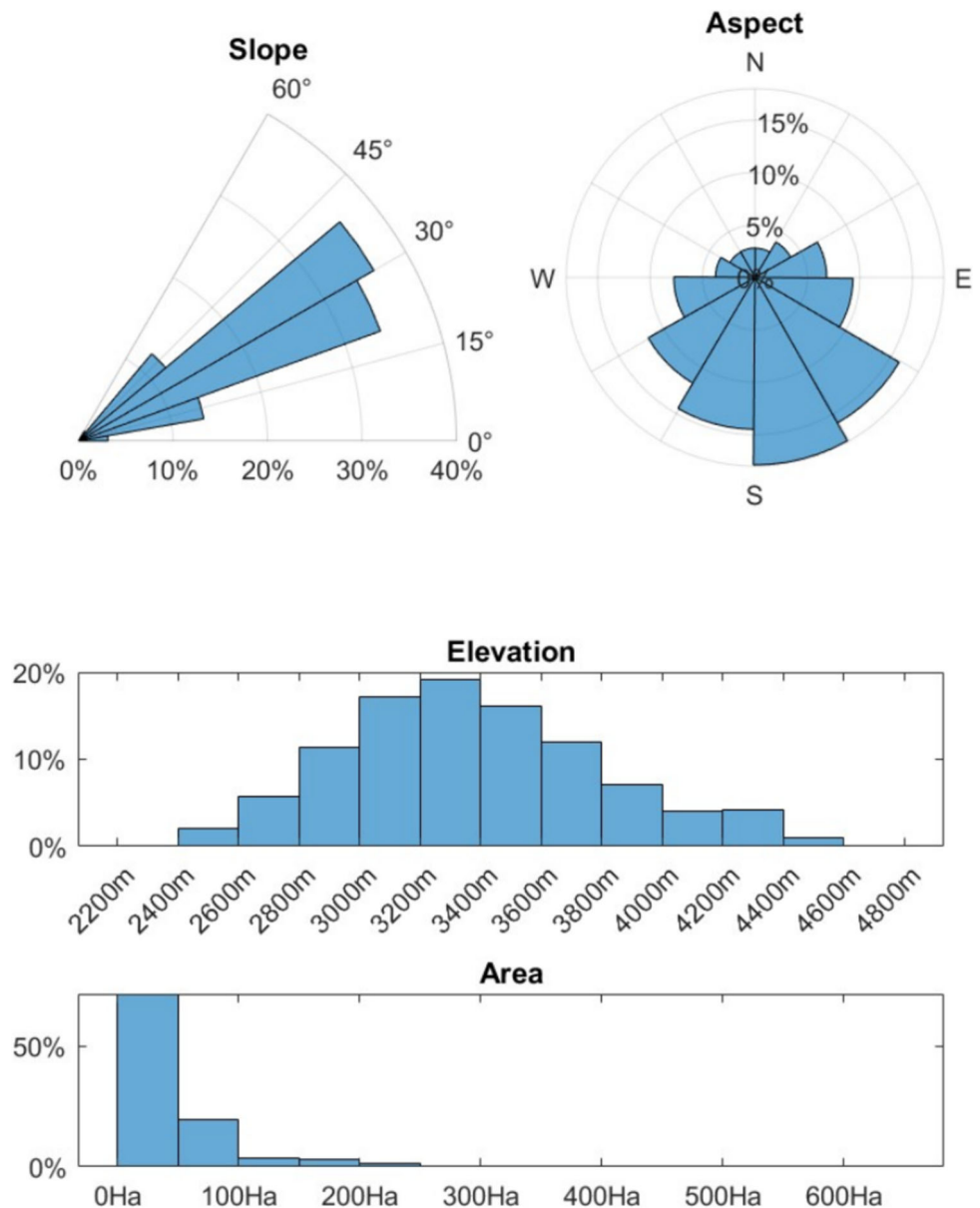


Fig. 1 A perspective view of the Bandipore-Gurej sector (study area). The thick black line feature represents the highway. There are 230 major avalanche paths (marked by black polygons) affecting the highway, lat-

eral roads, and tracks. A snow-meteorological observatory exists near Kanzalwan town. The shaded box in right-lower sub-image (white background) represents the study area on a wider map

Fig. 2 Distributions of terrain features of 230 representative avalanche paths of Bandipur-Gurez sector. The analysis is based upon counting of pixels representing the avalanche paths using ASTER Global Digital Elevation Model (30-m spatial resolution)



elevation-related sunshine effects, which become more evident in the later season. Therefore, constraining the analysis to the mere accumulation season, shortcomings attributed to a single station are mitigated to a large extent. Moreover, as mentioned by Pozdnoukhov et al. [6] and Kaur et al. [38], in practice, data describing meteorological (Class III) factors generalize over a larger region and are more commonly used in avalanche forecasting at the local and regional levels.

2.2 Feature Set

An instance of the 40-dimensional data represents the accumulated snow-meteorological condition at 08:30 IST (GMT+05:30) on a given day. The feature set as described

in Table 1 comprises of various avalanche contributing features as recommended in literature by experts. Features have been grouped as per their common physical nature. Feature set particularly includes the snow surface temperature and its hyperbolic tangent transformation as suggested by Pozdnukhov et al. [8] to obtain an increased sensitivity when approaching the melting point (0 °C). To take into account the temporal changes of these two features (a crucial piece of information on the evolution of snowpack stability), gradients over up to two previous days were also included as it has been felt by avalanche practitioners that the avalanches may result due to snow-meteorological perturbations occurred over past many days. A similar procedure was followed for the air temperature also. Water equivalent of fresh snow, water equivalent of aggregated seasonal snow,

Table 1 Details of 40 snow-meteorological features explored for their significance in avalanche formation

Feature Group	Feature Code	Feature Name	Feature Description	Unit	Class
Temperatures	F1	Air_Temp	Current day air temperature	°C	III
	F2	Air_T_Grad	Air temperature change from previous day	°C	III
	F3	Elaborate_Air_T	Hyperbolic tangent transform of air temperature		III
	F4	EAirT_Grad	Current elaborate air temperature minus previous day elaborate air temperature		III
	F5	EAirT_Grad_2	Current elaborate air temperature minus 2 days ago elaborate air temperature		III
	F6	Max_Temp	Maximum air temperature of past 24 h	°C	III
	F7	Min_Temp	Minimum air temperature of past 24 h	°C	III
	F8	Snow_Temp	Snow surface temperature	°C	II
	F9	Snow_T_Grad	Snow surface temperature change from previous day	°C	II
	F10	Elaborate_Snow_T	Hyperbolic tangent transform of snow temperature		II
	F11	ESnowT_Grad	Current elaborate snow temperature minus previous day elaborate snow temperature		II
	F12	ESnowT_Grad_2	Current elaborate snow temperature minus 2 days ago elaborate snow temperature		
Fresh snow	F13	Fresh_Snow	Fresh snow of last night	m	III
	F14	F_Snow_WEQ	Water equivalent of fresh snow of last night	m	III
	F15	FS24	Cumulative fresh snow of last 24 h	m	III
	F16	FS48	Cumulative fresh snow of last 48 h	m	III
	F17	FS72	Cumulative fresh snow of last 72 h	m	III
Seasonal snow	F18	Total_Snow	Cumulative seasonal snow amount	m	III
	F19	T_Snow_WEQ	Water equivalent of cumulative seasonal snow amount	m	III
Rainfall	F20	Rainfall	Rainfall during last night	m	III
	F21	Rain24	Cumulative rainfall of last 24 h	m	III
	F22	Rain48	Cumulative rainfall of last 48 h	m	III
	F23	Rain72	Cumulative rainfall of last 72 h	m	III
Humidity	F24	RH	Relative humidity	%	III
Snowpack	F25	Snow_Pent	Free ram penetration at snow surface	m	II
Radiation	F26	Sunshine	Sunshine during last 24 h	min	III
Wind	F27	Wind_Speed	Current wind speed	m/s	III
	F28	Wind_Direction	Current wind direction	°	III
	F29	Avg_Wind	Averaged wind speed of last night	m/s	III
	F30	Avg_Wind - 1	Averaged wind speed of previous night	m/s	III
	F31	Avg_Wind - 2	Averaged wind speed of pre-previous night	m/s	III
	F32	Wind24	Averaged wind speed of past 24 h	m/s	III
	F33	Wind48	Averaged wind speed of past 48 h	m/s	III
	F34	Wind96	Averaged wind speed of past 96 h	m/s	III
	F35	Cos_Wind_Dir	Cosine transform of the wind direction		III
	F36	Sin_Wind_Dir	Sine transform of the wind direction		III
	F37	Sin_Wind_Dir - 1	Sine transform of the previous day wind direction		III
	F38	Cos_Wind_Dir - 1	Cosine transform of the previous day wind direction		III
	F39	Sin_Wind_Dir - 2	Sine transform of the pre-previous day wind direction		III
	F40	Cos_Wind_Dir - 2	Cosine transform of the pre-previous day wind direction		III

and sine and cosine transformations of wind direction were also included following Pozdnukhov et al. [8]. Again, to represent the conditions of past few days individually as well as cumulatively, set of features from previous 1 to 2 days and some unified features were also considered in the manner adopted by Pozdnukhov et al. [8]. Features from previous day and pre-previous day are Avg_Wind-1, Avg_Wind-2, Sin_Wind_Dir-1, Cos_Wind_Dir-1, Sin_Wind_Dir-2, and Cos_Wind_Dir-2, whereas snowfall, rainfall, and wind speed features have been unified for up to past 96 h (e.g., cumulative fresh snow was extracted by unifying features from past 24, 48, and 72 h in order to exhibit snow accumulation, compaction, and abscission).

2.3 Avalanche Observations

The other very important data used to conduct this study is avalanche occurrence observations from the study area. These observations were needed to label each day as “avalanche day” or “no-avalanche day.” Indian Army personnel deployed at the border outposts as well as local villagers report these occurrences. The observation details generally include location, date and time of occurrence, the altitude and slope aspect of avalanche path, nature of occurrence (natural or human-triggered), type of avalanche (dry, wet, powder, etc.), size of debris, and damage caused, if any. However, many of these properties may be estimated or not available at all due to terrain and weather related difficulties. Efforts are also made to extract details (exact location, fracture area estimation, extend of debris, etc.) from available satellite imagery for the relevant area and of the period

immediately after the occurrence. Nevertheless, it is very much likely that many avalanche occurrences might have gone unnoticed due to challenges of data observations in difficult mountain terrains and unfavorable weather conditions. Such inherent uncertainties and lack of high reliability in avalanche data as well as its effect on precision of avalanche modeling have been recognized in many earlier studies such as Bonnefoy-Demongeot et al. [39], Pérez-Guillén et al. [40], and Mayer et al. [14]. Based upon the available information, it can be inferred that avalanche activity period in BG sector is restricted mainly to typical winter months in this part of the globe, i.e., November of a year to April of next year.

2.4 Data Preparation and Pre-processing

For the present study, we considered 40-dimensional snow-meteorological data and avalanche observations as described in previous sub-sections for 12 consecutive winter-seasons (2011/2012 to 2022/2023). Only those instances were considered which had data available for all the 40 features. An instance was labeled as “avalanche day” (or “no-avalanche day” otherwise), if at least one avalanche was reported in the study area on the date of instance. In total, 1107 instances were found to be complete in all respect, out of which 187 instances were labeled as “avalanche day” and 920 instances were labeled as “no-avalanche day.” Figure 3 depicts the summary statistics of the 40-dimensional dataset used in this study (Table 1) through boxplotting.

Boxplot representations exhibit that many features have skewed distributions. Particularly, features belonging to the fresh snow group (F13, F14, F15, F16, and F17) and rain-

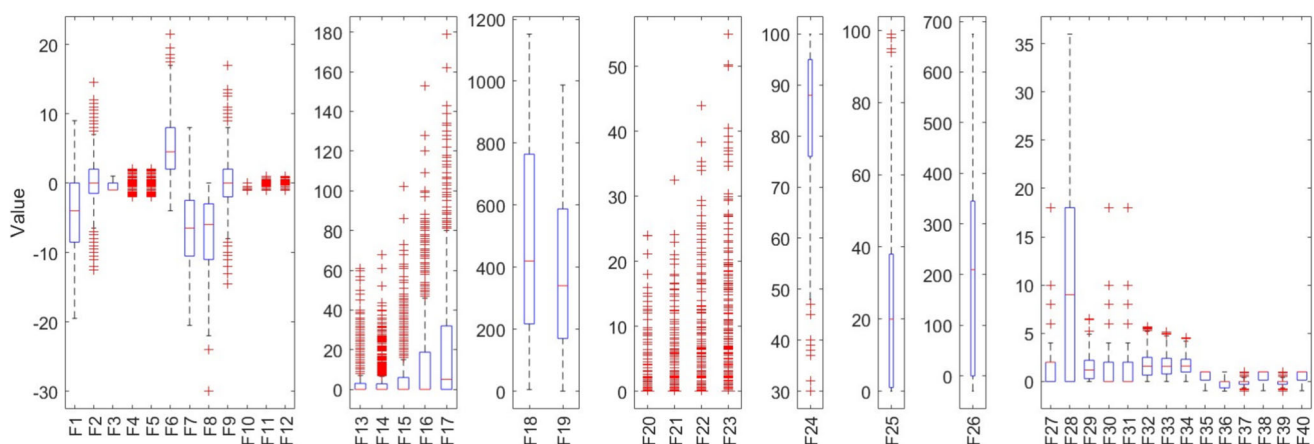


Fig. 3 Summary statistics of 40-dimensional snow-meteorological dataset used in the study. Features have been grouped according to their shared physical nature (refer to Table 1). *x*-axis accounts for features (F1 to F40), and *y*-axis holds their values. Boxes' ends represent inter quartile ranges (1st to 3rd quartiles), middle horizontal bars in each

box denotes the median (2nd quartile), and whiskers (vertical lines with horizontal bars) represent the highest and lowest data value within 1.5 times the inter quartile range above the 3rd or below the 1st quartile, respectively. The data points above or below the whiskers are the possible outliers

fall group (F20, F21, F22, and F23) are extremely skewed as majority of the data values are zero while remaining values are positive. Besides, features belonging to transformations of temperatures (F3, F4, F5, F10, F11, and F12) and transformations of wind direction (F35, F36, F37, F38, F39, and F40), though symmetrically distributed, also have majority of data values as zero. Consequently, all the quartiles and data range indicators (within 1.5 times of inter-quartile range above the 3rd or below the 1st quartile) for most of these features have converged to zero value while the non-zero values are represented as outliers. It is believed that these outliers hold high significance in avalanche formation process. Further, the month-wise distribution of 187 “avalanche day” instances used in this study is presented in Fig. 4.

As a data pre-processing step, we applied z -score normalization over the snow-meteorological dataset so as to rescale each feature with zero mean and unit standard deviation. Further, the entire dataset was random split into two equal parts where one part (50%) was retained for feature subset selection experiments and model training while remaining (50%) data was used for model testing for operational performance estimation.

3 Methods

This section describes the various algorithms, validation methods, and the employed methodology to conduct this study.

3.1 SVM-RFE Algorithm for Feature Selection

Recursive feature elimination (RFE) is a gold standard of wrapper feature selection as RFE is much more robust to data overfitting than other methods [29]. On the other hand, support vector machine (SVM) is also an effective tool with significant merits and discovers informative patterns from the

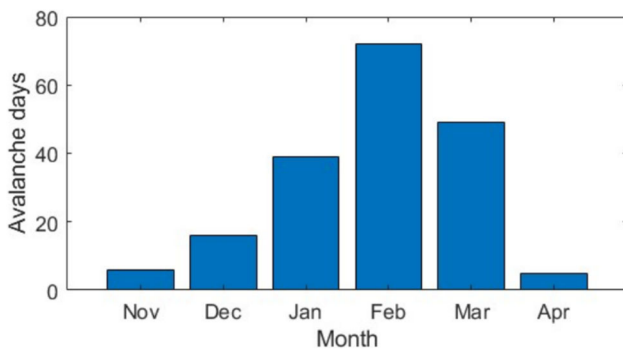


Fig. 4 The month-wise distribution of 187 “avalanche day” instances in the dataset used in the study of period from 2011/2012 to 2022/2023 winter seasons

input data [41, 42]. SVM-RFE, initially proposed for gene selection for cancer classification [29], is a backward feature elimination technique which uses the weight vector of the hyperplane constructed by the samples on the margin, i.e., support vectors as ranking criterion. Final output of this technique is a ranked list with features ordered according to their relevance. SVM-RFE improves features selection based on feature ranking by eliminating the orthogonality assumptions of correlation methods [29]. To incorporate this feature selection algorithm in the current study, we utilized the core notion of SVM-RFE technique, i.e., features with least influence on the decision surface of the SVM classifier are excluded in an iterative manner. Thus, at each iteration of algorithm, feature with lowest weight is dropped. Dropping or elimination process continues until the input feature set is empty. Subsets selected at each iteration are analyzed and compared according to a performance evaluation criterion to select the optimal feature subset. Pseudo-code of SVM-RFE algorithm as proposed by Guyon et al. [29] is presented in Fig. 5

Inputs:

Training examples

$$X_0 = [x_1, x_2, x_3, \dots, x_k, \dots, x_l]^T$$

Class labels

$$y = [y_1, y_2, y_3, \dots, y_k, \dots, y_l]^T$$

Initialize:

Subset of surviving features

$$s = [1, 2, 3, \dots, n]$$

Feature ranked list

$$r = []$$

Repeat until $s = []$

Restrict training examples to good feature indices

$$X = X_0(:, s)$$

Train the classifier

$$\alpha = \text{SVM-train}(X, y)$$

Compute the weight vector of dimension $\text{length}(s)$

$$w = \sum_k \alpha_k y_k \cdot x_k$$

Compute the ranking criteria

$$c_i = w_i^2, \text{ for all } i$$

Find the feature with smallest ranking criterion

$$f = \text{argmin}(c)$$

Update feature ranked list

$$r = [s(f), r]$$

Eliminate the feature with smallest ranking criterion

$$s = s(1 : f - 1, f + 1 : \text{length}(s))$$

Output:

Feature ranked list r

Fig. 5 Pseudo-code for SVM-RFE algorithm

3.2 The Classifiers: Support Vector Machine and Random Forest

Support vector machine (SVM) as proposed by Vapnik [42] is a maximum margin technique, utilized for classification and regression. SVM can deal with linear and nonlinear problems. Training of SVM comprises discovering a maximum margin decision surface, also referred as hyperplane, to categorize negative class labels from positive class labels. This maximum margin property of SVM will provide the best generalization and lowest validation error [8]. A subset of input data called as support vectors is responsible for generating hyperplane, while, considering the nonlinear problems, a quadratic programming kernel trick is used which transforms the data to higher dimension. A kernel function is defined as follows: $K(x, x') = \psi(x) \cdot \psi(x')$. Commonly employed kernel functions are linear, polynomial, and radial basis function (rbf) kernel [8]. Support vector machine has widely gain popularity for its efficiency and high generalization abilities in the domain of avalanche forecasting [6, 8, 10, 13, 20]. In this study, we deployed SVM classifier with linear kernel in association with already established feature selection technique (SVM-RFE) for avalanche forecasting over the study area in Indian Himalayas. Linear kernel is defined as follows: $K_{linear}(x, x') = x \cdot x'$, where x and x' are the input vectors. An improvement in SVM, known as soft-margin SVM [8, 42], was formulated for correct classification of noisy real world data. Now, motivation was not to make zero classification mistakes, but to make as few mistakes as possible. Hence, C (regularization parameter) was introduced. For operational purpose of SVM, C is the hyperparameter that is needed to be optimized, which controls the trade-off between the maximization of the margin and the permitted training errors [8].

Random forest [43] is an ensemble of decision trees belonging to the category of supervised learning algorithms. A decision tree follows an inverted tree like structure for classifying the query. The classification starts from root node. Each tree node is a step of classification process, and non-terminal node (internal node) represents a test condition. Future steps are represented by outgoing arrows from non-terminal node owing to the outcome of test condition. These outgoing arrows are further connected to nodes which are termed as children nodes or child nodes. Those nodes with no outgoing arrow and no child node are referred as the leaf node (terminal node). To classify a query, the test condition is applied to the root node, and depending on the test condition, a child node is selected. If this child node is not the leaf node, then this process continues to subsequent child nodes till leaf node reaches. Final classification decision is returned by the leaf node. Training process of RF involves bootstrap aggrega-

tion (also called bagging) technique that trains multiple decision trees on different parts of the same training set and ensures no correlation between individual trees. Final classification prediction is done by majority voting process from the ensemble of decision trees. The splitting on the nodes of the decision tree is done using Gini Index to regulate which of the branches have more occurrence probability. In RF, two hyper parameters are needed to be optimized, i.e., N (number of decision trees) and D (maximum tree depth). With a greater “number of decision trees,” an increment in classifier’s performance is observed with more stable predictions and “maximum tree depth” helps in capturing the more and more information from the data.

3.3 Validation Methods

For model validation, i.e., performance evaluation, few highly recommended measures have been used, i.e., receiver operating characteristics (ROC) curve and the area under ROC curve (AUC). The following is a brief description of these measures.

- Receiver operating characteristics (ROC) curve: This is a prominent technique for visualizing, organizing, and selecting classifiers based on their performance [44–46]. It is a unit square two-dimensional plot between true positive rate (TPR) on y-axis and false positive rate (FPR) on x-axis with varying thresholds. Flach [47] presented a detailed analysis on ROC curves and addressed many problems such as (1) determining a decision threshold that minimizes error rate or misclassification cost under given class and cost distributions, (2) identifying regions where one classifier outperforms another, (3) identifying regions where a classifier performs worse, and (4) obtaining calibrated estimates of the class posterior.

- Area under ROC curve (AUC): This is an efficient threshold independent measure for estimating the performance of classifiers in a single scalar value, i.e., more the AUC value and better the classification performance. Since the AUC is a portion of the area of the unit square, its value will always be between 0 and 1 [44–46]. AUC has a very useful statistical interpretation: it is the expectation that a (uniformly) randomly drawn positive receives a higher score than a randomly drawn negative. Of late, AUC has been employed in many studies for performance estimation [13, 44–49] and is called as the measurement of classifier’s skills based on the surface [45, 46, 48].

ROC curve and AUC are the metrics that are independent of data skewness. Further, AUC is more discriminative than other simple metrics such as accuracy (also known as “proportion of correct classifications”), F -measure, precision, and recall [31, 45, 46, 48]. ROC curve and its respective

AUC score secure a threshold independent estimation of a classifier performance in contrast with accuracy. Due to its framework of plotting true positive rate against false positive rate, it provides a sort of normalization for both by observed number of positive and negative illustrations. This normalization assures that ROC-AUC remains unaffected from data skewness, as it analyses model's capability to classify the binary events rather than depending upon the absolute counts [44, 50].

4 Study Plan

The plan followed to conduct this study is presented in Fig. 6 with following major steps: (1) preparation of 40-dimensional dataset and its pre-processing, i.e., z-score normalization alongwith its labeling in terms of “avalanche day” or “no-avalanche day”; (2) dataset random splitting into training and testing data subsets respectively in 1:1 ratio; (3) feature subset selection algorithm implementation and model learning using training data where SVM-RFE algorithm was implemented to find out the significant avalanche contribut-

ing feature subsets and then classifier was trained using the same feature subsets with training performance measured in terms of five-fold cross-validated AUC; (4) the feature subset corresponding to highest AUC from previous step was selected as optimal subset; and (5) classifier's generalization performance was recorded on testing data filtered with optimal feature subset in terms of ROC curves and the corresponding AUC values.

5 Results and Discussion

5.1 Feature Selection

The effects of feature selection step (through SVM-RFE algorithm) on the performances of SVM and RF classifiers are presented in Fig. 7. Cross-validated AUC values determined over training data were used to evaluate the performance of classifiers with different feature subsets corresponding to each iteration of features selection algorithm.

The SVM classifier demonstrated the best performance (AUC=0.7946) at 29th iteration, i.e., with 12 features,

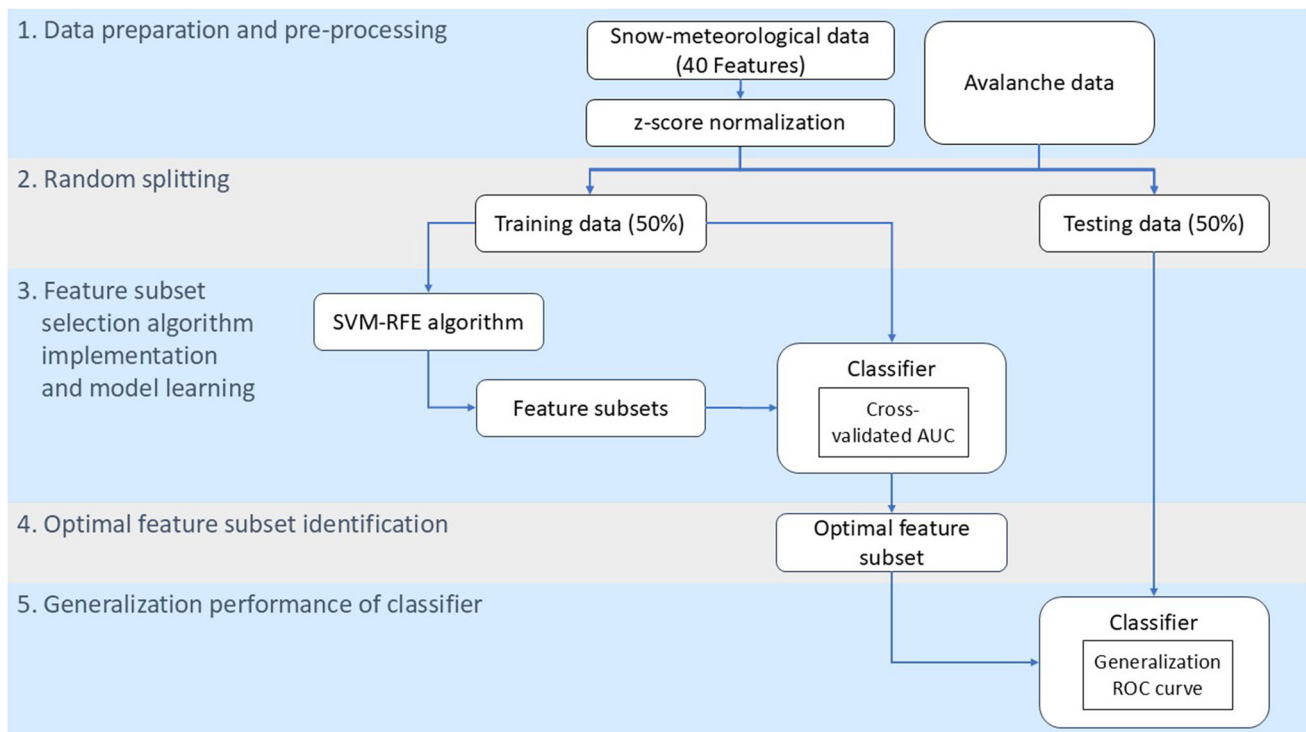
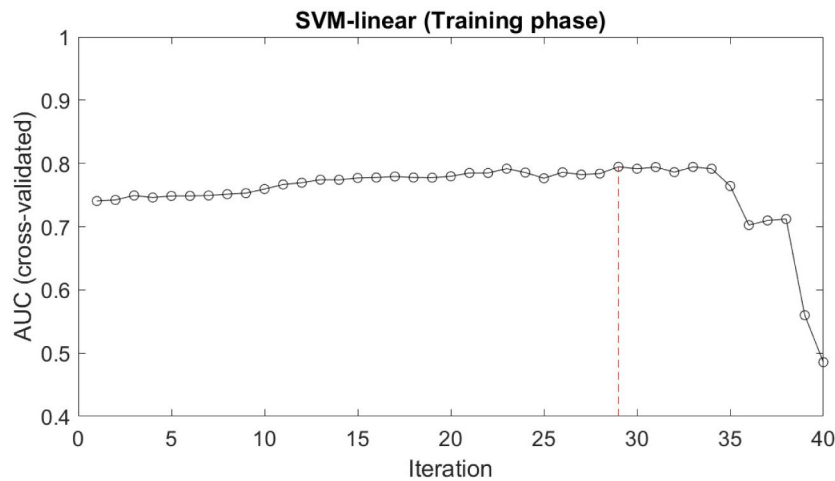
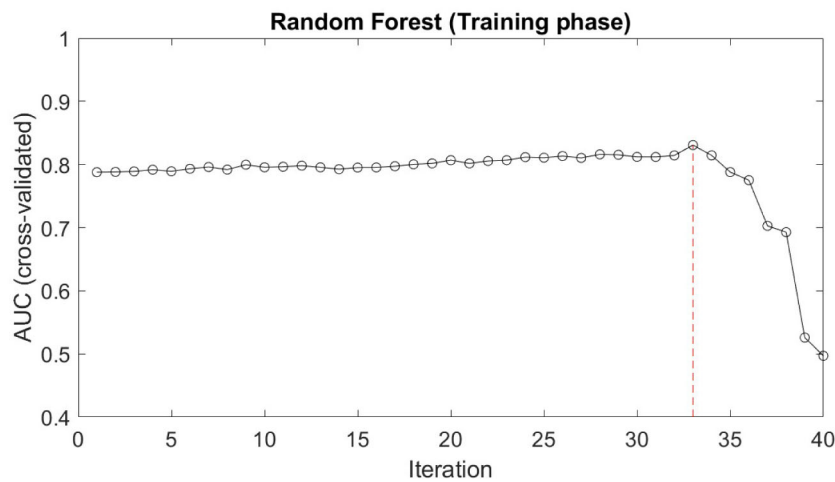


Fig. 6 The study implementation plan



(a) Optimal performance obtained at 29th iteration.



(b) Optimal performance obtained at 33rd iteration.

Fig. 7 The performance trends of classifiers, i.e., **a** SVM with linear kernel and **b** RF, presented in terms of cross-validated AUC with different feature subsets corresponding to each iteration of feature selection

algorithm (starting with all 40 features and dropping a feature with each subsequent iteration). Optimal performance for each classifier is marked with red vertical dotted line

whereas, in case of RF, the best performance (AUC=0.8307) was achieved at 33rd iteration with eight features. In both the cases, these AUC values are noticeably improved as compared to that achieved with all the 40 features, i.e., the performances at first iteration (0.7405 for SVM and 0.7878 for RF). A significant drop in performances has been observed from 35th iteration onwards for both the classifiers, implying that there is significant information loss with every feature dropped at 35th iteration onwards. Overall, these results imply that the feature subsets used from about 26th iteration (i.e., 15 features) onwards and up to 34th iteration

(with 7 features) may be considered significant as within this region, both the classifiers have consistently offered higher performance (with cross-validated AUC of above 0.78 for SVM and above 0.81 for RF). The features subsets used by classifier algorithms at 26th iteration and 34th iteration are shown in Table 2.

Selected significant features represent temperatures (F6, F7), fresh snow (F13, F15, F17), seasonal snow (F18, F19), rainfall (F22, F23), snowpack (F25), radiation (F26), and wind (F32, F33, F39, F40) groups. Among these, while “Snow_Pent” (F25) belongs to Class II, the others belong

Table 2 Significant avalanche contributing features subsets obtained by SVM-RFE algorithm at 26th and 34th iterations

Feature rank	Iteration = 26		Iteration = 34	
	Feature code	Feature	Feature code	Feature
1	F23	Rain72	F23	Rain72
2	F22	Rain48	F15	FS24
3	F18	Total_Snow	F22	Rain48
4	F19	T_Snow_WEQ	F13	Fresh_Snow
5	F17	FS72	F19	T_Snow_WEQ
6	F15	FS24	F18	Total_Snow
7	F6	Max_Temp	F17	FS72
8	F33	Wind48		
9	F13	Fresh_Snow		
10	F25	Snow_Pent		
11	F40	Cos_Wind_Dir - 2		
12	F32	Wind24		
13	F7	Min_Temp		
14	F39	Sin_Wind_Dir - 2		
15	F26	Sunshine		

to Class III. Though these features were selected in a purely data-driven way, their physical relevance to the formation of snow avalanches is well recognized by various avalanche experts [14, 35, 51–54]. Overall stability of the snowpack is dependent on amount of snowfall and temperature as noted by Choubin et al. [10] and Mayer et al. [14]. According to Strapazzon et al. [54], a variation in temperature will cause changes in the frequency, intensity, and types of snowfalls, according to elevation. Air temperature is one of the most important feature, as it affects snow accumulation and formation of weak layers within the snowpack [10]. Features related to wind play an important role in snow drifting and accumulation [6, 10, 14, 53]. As per the discussions of Schweizer et al. [53], the wind transforms the snow into a more brittle material and that wind-deposited snow layers are more prone to avalanching. Stability of the snowpack is affected by fresh snow and total snow [14, 53, 55]. According to Mayer et al. [14], the amount of new snow is known to be one of the critical indicators for avalanches and its level of effectiveness depends on several factors including thickness of precipitation, wind transported snow, and temperature. Conway and Raymond [56] presented a detailed case study on snow stability during rain in the central Cascade Mountains, Washington, USA. Following the onset of rain, three evolutionary regimes of snow behavior were identified: immediate avalanching, delayed avalanching, and return to stability. Perla [57] presented a case study of avalanching at Alta, Utah, and Loveland-Berthod, Colorado, and concluded

that the features: temperature, wind, snow, sunshine, rain-fall, and relative humidity and few of their derivatives are the most significant avalanche contributing features. The impact of feature groups selected through this study on avalanching is summarized in Table 3.

Another interesting aspect to note here is that many of the selected features subset (particularly F17, F22, F23, F33, F39, and F40) represent information up to 2 to 3 days before the forecast day. Here, it can be inferred that in this study area, avalanching conditions ripe over a longer period of time and not necessarily as a result of rapid meteorological changes. In Table 2, although significant features have been placed as per their respective ranks determined by SVM-RFE algorithm (according to their corresponding weights from the decision functions constructed), in practice, the entire feature subset is considered a unified entity as RFE ranking is a feature subset ranking and the nested feature subsets contain complementary features not necessarily individually most relevant. Only taken together, the features of a subset are optimal in some sense [29]. Still, it is interesting to note that 6 out of top 7 features in the two subsets are common. Further, during the entire RFE exercise of 40 iterations, how the various features fared in terms of their individual ranks within the subsets and how long a particular feature survived were tracked, and these characteristics have been summarized in Fig. 8. Here, it is interesting to note that the 12 features of the optimal feature subset corresponding to iteration 29 (refer Table 2) almost always had higher ranks (third quartile of ranks distri-

Table 3 Impacts of significant avalanche contributing features as selected by SVM-RFE algorithm

Feature group	Impact	References
Seasonal snow	The amount of snow across the region is directly related to avalanche occurrence probability	[14, 51, 52, 57]
Fresh snow	For large (catastrophic), new snow avalanches, fresh snow is the strongest forecasting parameter. Fresh Snow adds a new snow layer to the snowpack and increased snowfall at high elevations can lead to peaks in avalanche activity	[51, 52, 52, 57], [14, 53, 55]
Wind	Wind speed leads to snow redistribution and drifting. Wind with high speed carry snow rapidly and create sturdy drifts that append strain to the snowpack Wind direction is a significant feature in avalanche forecasting as it plays an important role in snow accumulation, redistribution and avalanche occurrence	[51–53, 57], [14] [6, 51, 53, 57], [10]
Temperature	The temperature within the snowpack contributes to the overall stability of snowy areas. Upper layers of snow can get seriously weakened by sudden rise in temperature causing snow melt within the snowpack, which can increase the chances of snow avalanches Gradual rise in temperature during snowfall can form cohesive layers of snow over weaker layers which triggers avalanches	[10, 51, 53, 57], [13, 54]
Radiation	The sunshine on snowpack, makes them loosened, causes instability in snowpack and this instability leads to snow avalanche. These kind of avalanches typically originates from a single point and progressively widens while descending through steep slopes	[51, 57]
Rainfall	Rainfall on snow is a typical feature that plays an important role by causing snow-metamorphism which in turn creates a thin and weak snow layer triggering avalanches	[51, 52, 56]

bution is less than 20) indicating that these features remained among top 20 features almost throughout the exercise.

5.2 Validation

By using the optimal feature subsets, the results of generalized performances of two classifiers over testing data are presented in Fig. 9 in terms of ROC curves. For comparison, the ROC curves corresponding to generalization

performances without feature selection (i.e., with all 40 features) are also included for respective classifiers. For reference, theoretical curves corresponding to random forecast (no skill) are also shown. For both the classifiers, the generalization performances with feature selection are either matching or exceeding their respective performances without feature selection in terms of AUC (for SVM, 0.6041 in both the cases while for RF, 0.6081 with feature selection and 0.6047 without feature selection, respectively). These

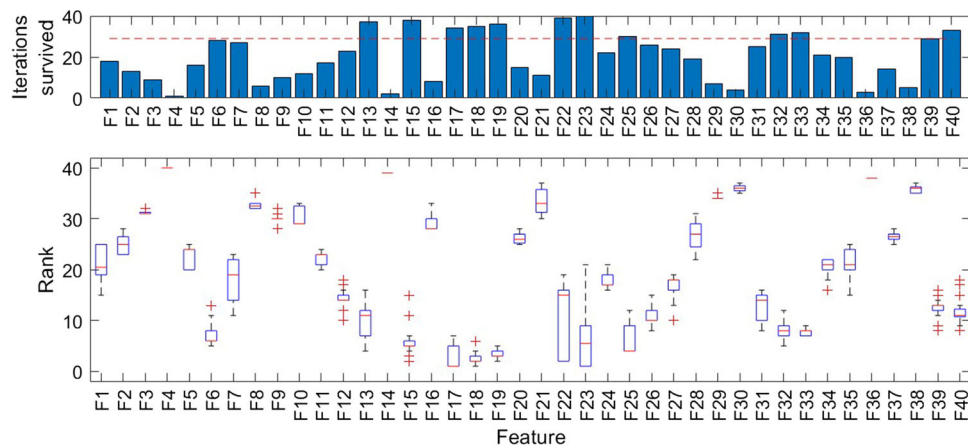


Fig. 8 RFE experiment tracking through the 40 iterations of SVM-RFE algorithm. Box plots (lower plot) represent distribution of ranks of various features through the experiment, and bars (upper plot) represent

the last iteration numbers before corresponding feature gets dropped. The horizontal red dotted line represents the 29th iteration marking the optimal features subset for SVM-linear classifier

results validate the importance of role played by a good feature selection method for classification modeling as similar or better performance could be achieved with much lesser number of features for two conceptually varying classifiers.

5.3 Comparison with Previous Study

As per our knowledge, Pozdnoukhov et al.'s [6] study is the only other work available that is related to the application of feature selection methods for avalanche forecasting where SVM-RFE was deployed for feature selection from an initial feature set of 40 features that included standard and

expert features. It selected subsets of 8 to 16 features as most influencing. Though direct one-to-one comparison is difficult between the two studies due to different initial features sets as well as data from different climatic regions, broadly both the studies retained features representing both Class II (snowpack) and Class I (meteorological) information. Nevertheless, a distinctively contrasting result from these two studies is that while Pozdnoukhov et al. [6] selected features that largely represented the effect of rapid change in meteorological conditions (only 2 features out of 20 were retained that represented condition of 2 days before the forecast day), the present study selected 6 features out of 18 that represented the effect of meteorological change over last 2 to 3 days.

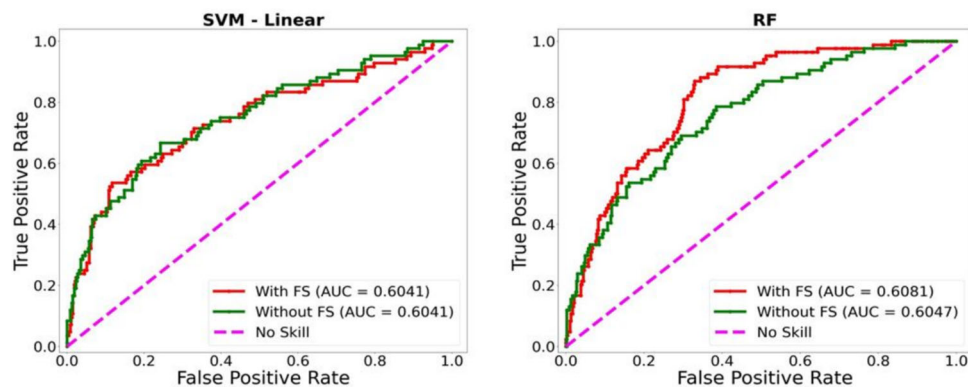


Fig. 9 Results of generalization performance for avalanche forecasting by two classifiers with feature selection (FS) in comparison to that without feature selection in terms of area under ROC curves. The SVM

and RF results with feature selection correspond to feature subsets used during 29th iteration and 33rd iteration, respectively. For reference, the theoretical curves representing random forecast (no skill) are also shown

6 Conclusions

Through a detailed investigation over data from Indian Himalaya, this study establishes that feature selection should be considered a mandatory prerequisite for an efficient avalanche forecast model building. The study suggested a subset comprising of 15 features on the liberal side and a subset comprising of 7 features on the stricter side and demonstrated consistent or improved performance of two theoretically varying but well-established machine learning-based classification models over the performance with 40 features. The almost identical performance results for the two classification algorithms establish the veracity of the results. Probably, it also signifies the high physical relevance of selected features with the avalanche formation process. For the study area of this work, the results have also stressed upon the features that represent the influence over the previous up to 3 days from the forecast day. Overall, the results of present study align with existing literature in terms of the nature of selected features for avalanche forecasting. This is the first detailed study of its kind on the use of any feature selection method for identifying significant feature for avalanche forecasting. As a future research direction, we recommend exploring other methods of feature selection as well as incorporation of data from other climatic regions for better understanding of nature of features influencing avalanches. Assimilation of spatially distributed and remote sensing based features is also suggested to augment the representativeness in diverse terrains.

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Data Availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of Interest The authors declare no competing interests.

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